

Obscuration de chiffres dans des images par translation de vecteurs latents

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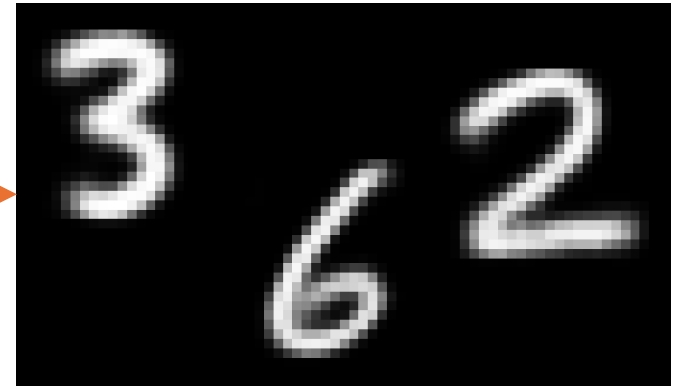
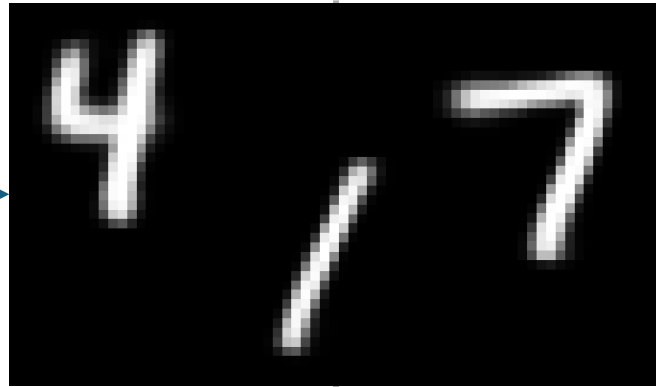
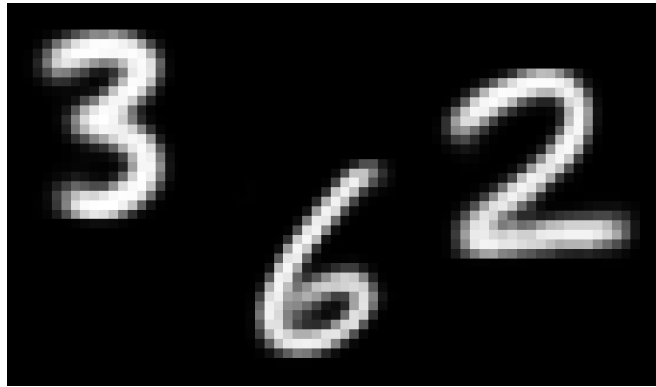
Contexte

- **Protection des données**, de la vie privée
- Importance de l'information dans les médias visuels
- Obscuration **réversible** à partir d'une clé
- Bonne **fidélité visuelle** et **invisible**



Solution

Obscuration



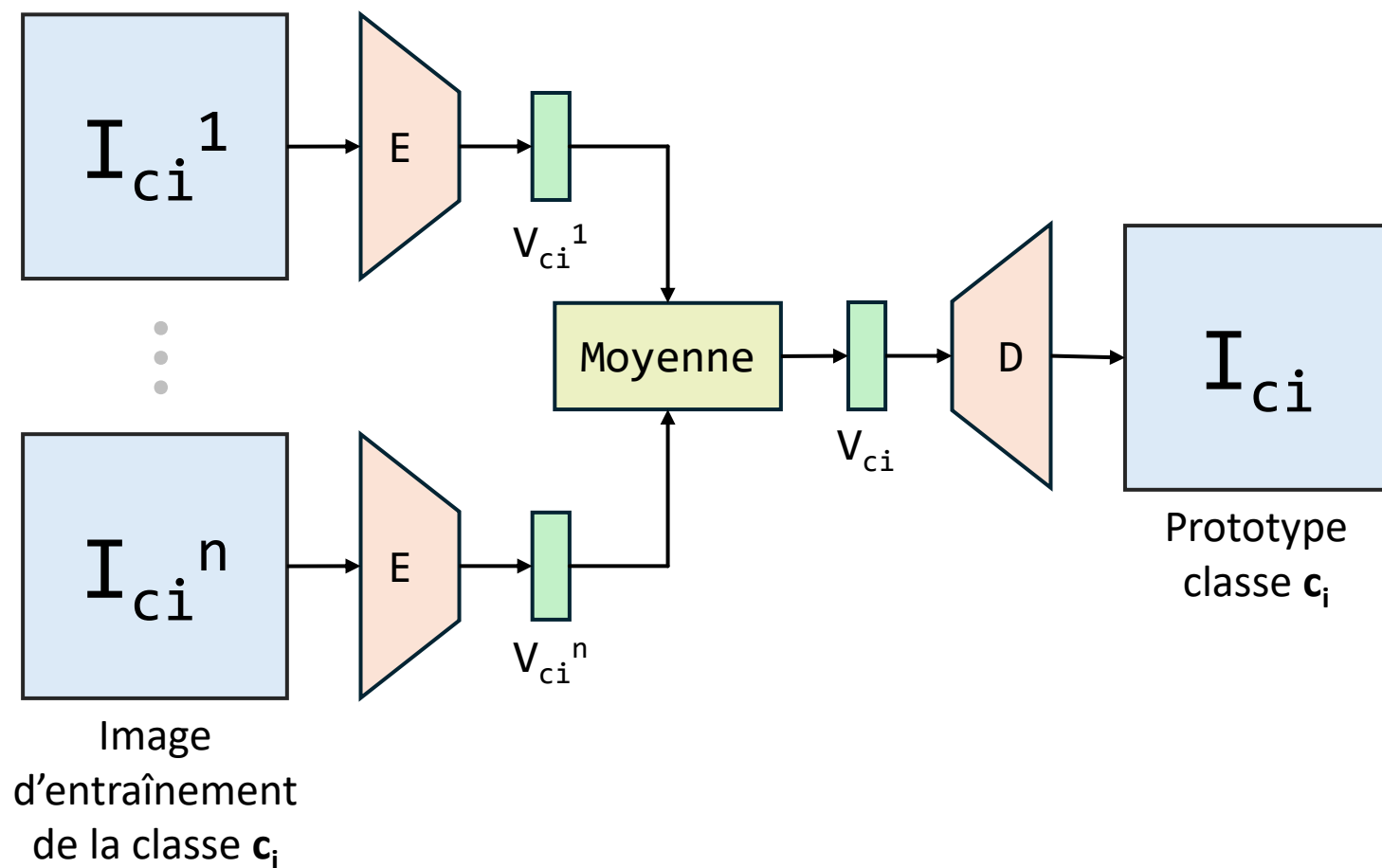
Désobscuration

Méthode

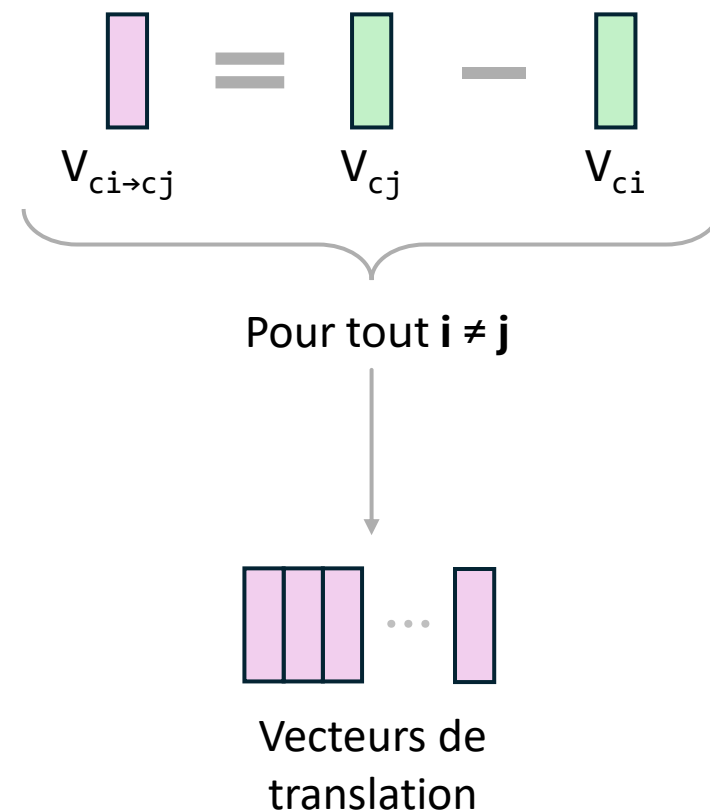
Méthode

Calcul des vecteurs de translation

Calcul du centroïde de la classe $\mathbf{c}_i$

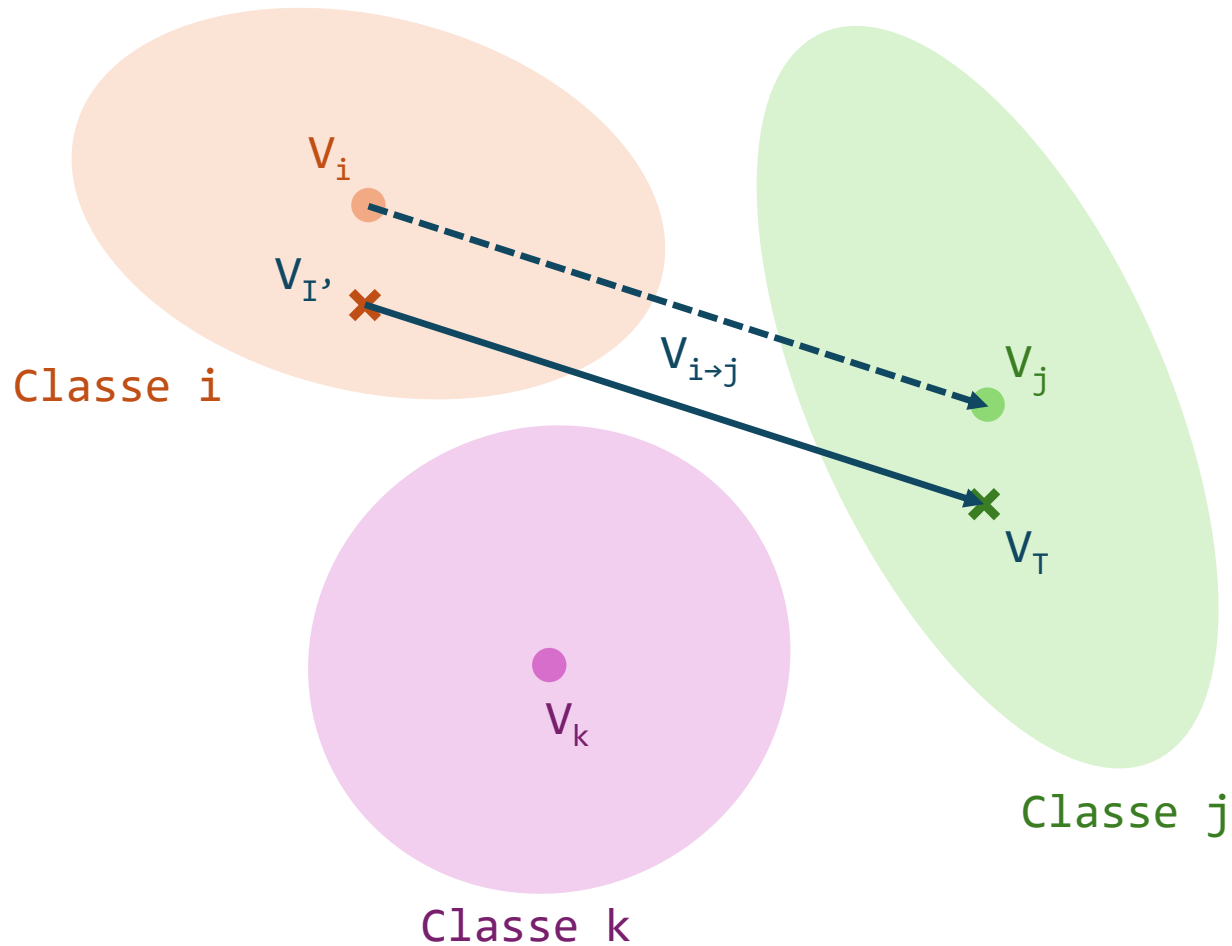


Calcul de la translation de la classe source $\mathbf{c}_i$ vers la classe cible $\mathbf{c}_j$



Méthode

Obscuration



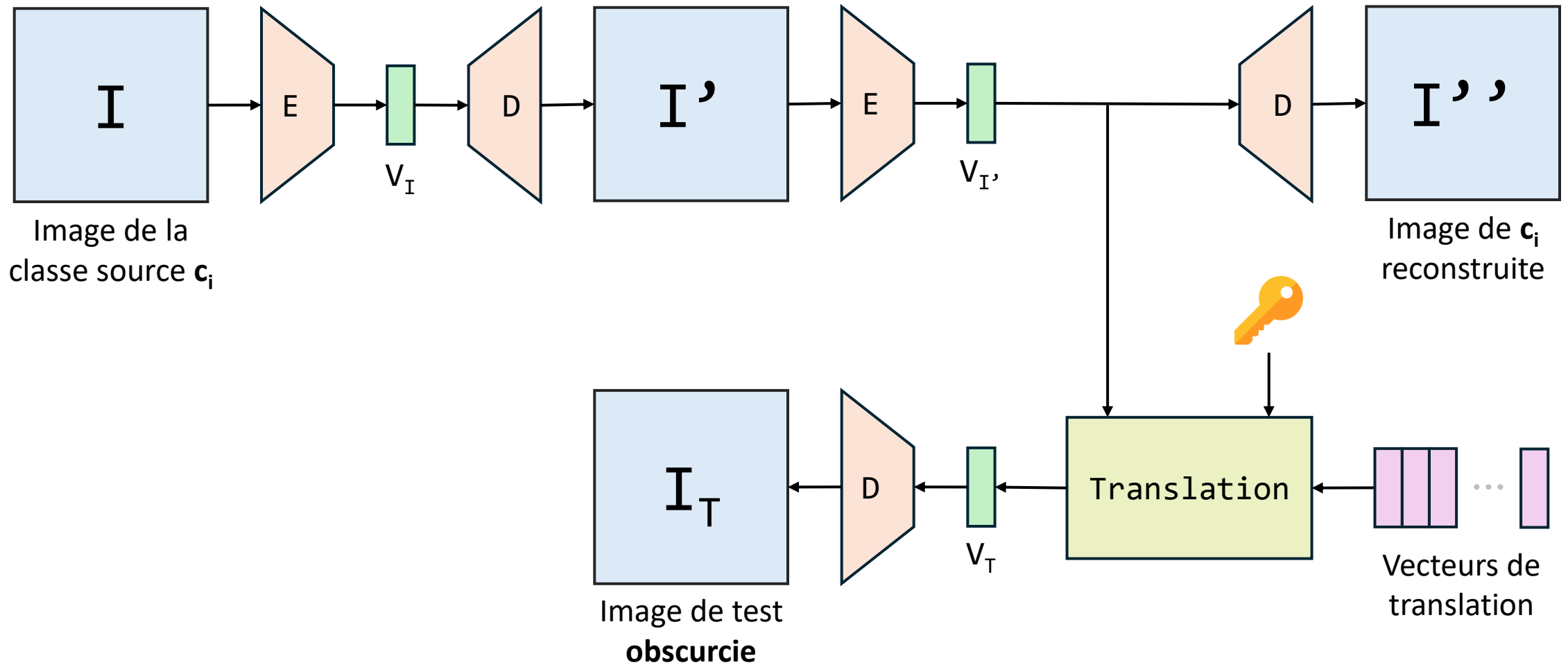
Formule de la translation de
la classe **i** vers la classe **j** :

$$V_T = V_{I'} + V_{i \rightarrow j}$$

Les classes source et cible
sont connues

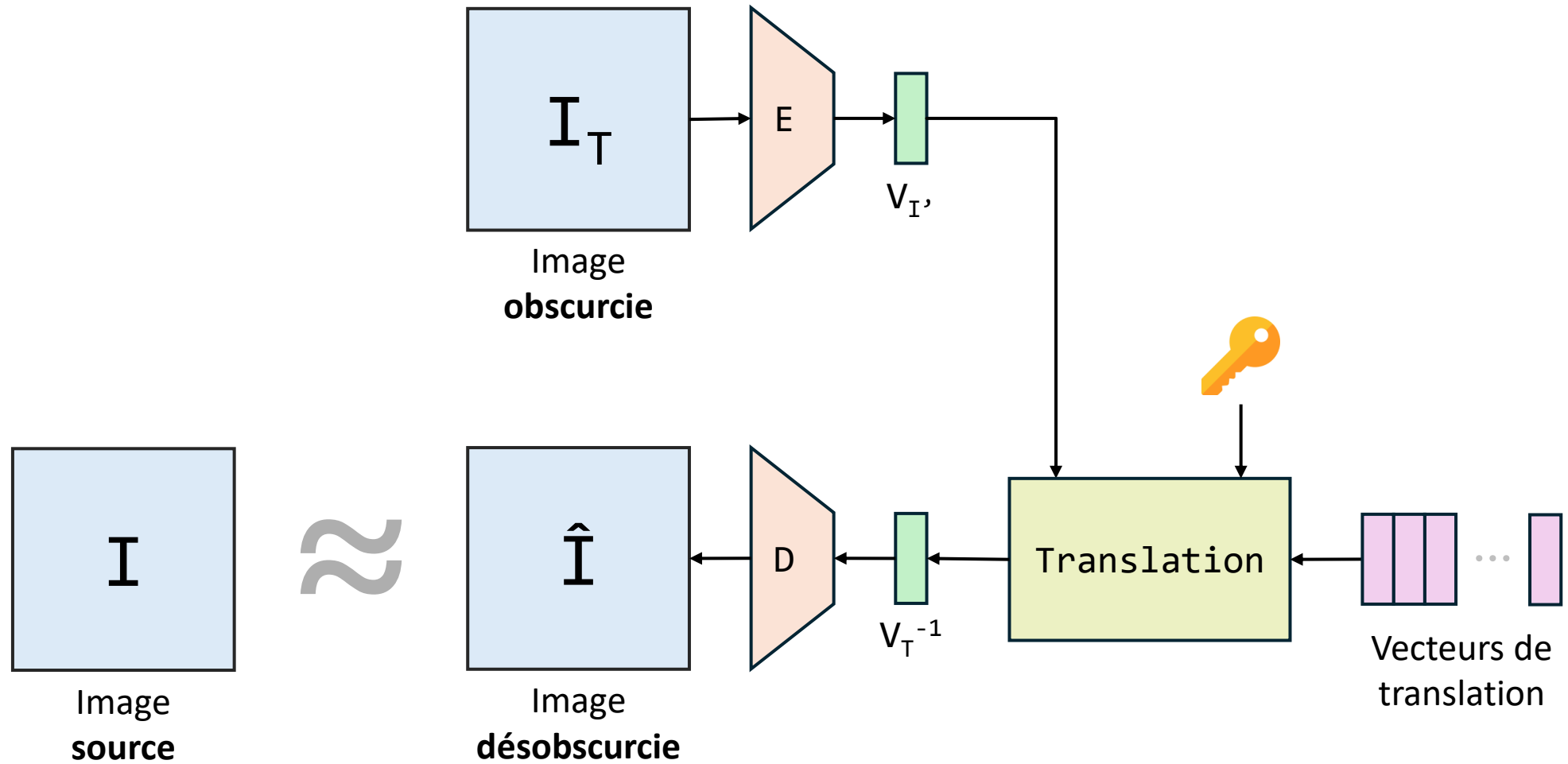
Méthode

Étape d'obscurisation



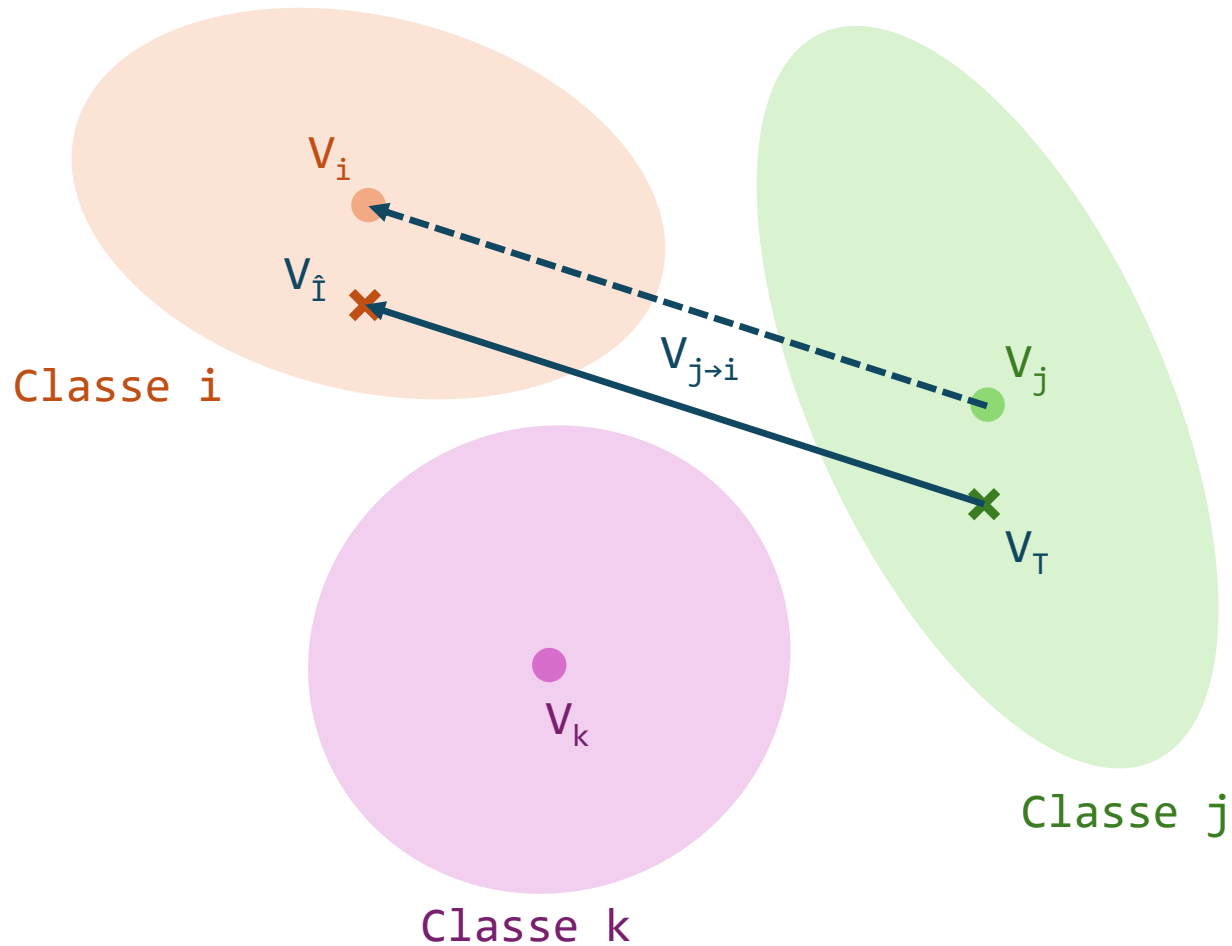
Méthode

Étape de désobscuration



Méthode

Désobscuration



Formule de la translation inverse
(classe j vers la classe i) :

$$V_T^{-1} = V_T + V_{j \rightarrow i}$$

Les classes source et cible sont
connues

Méthode

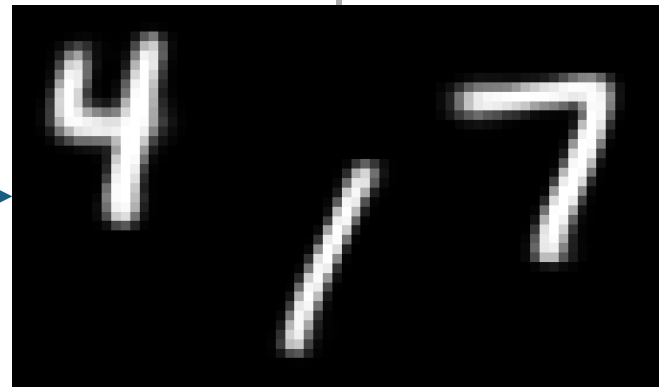
Évaluation de l'obscurité (classification)

Obscuration



Classifieur

3 6 2



Classifieur

4 1 7

Désobscuration



Classifieur

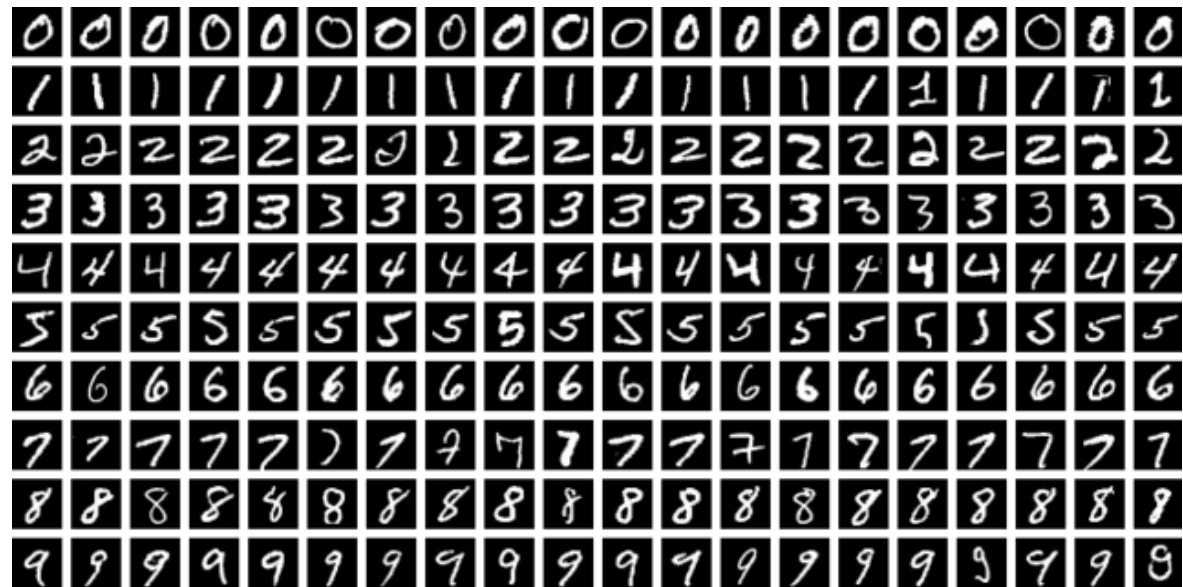
3 6 2

Résultats expérimentaux

Résultats expérimentaux

Dataset

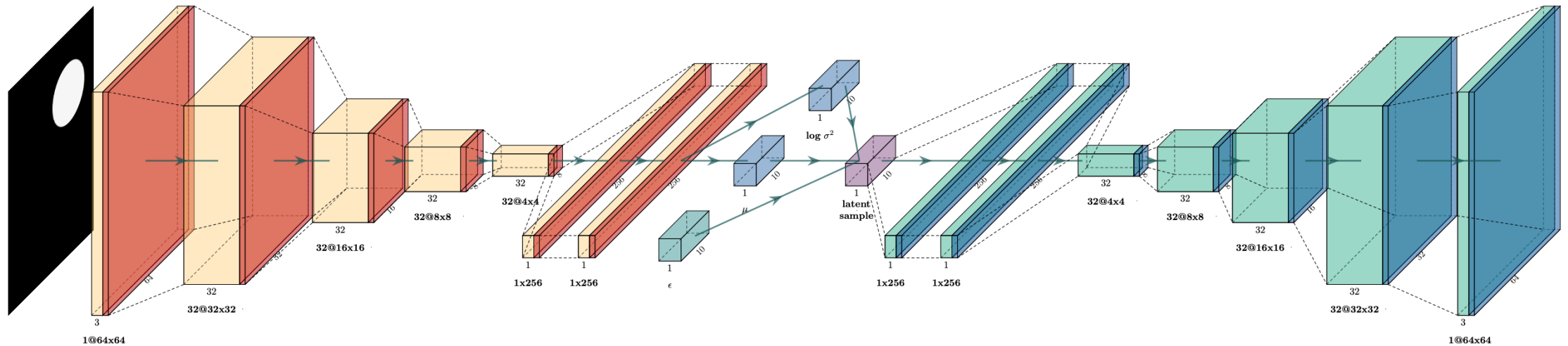
- Dataset d'images dessinées **MNIST**
- **60 000** images d'entraînement
- **10 000** images de test



Résultats expérimentaux

Modèles

VAE



- Autoencodeur variationnel
- Facteur β de démêlage
- Vecteur latent de taille **128**

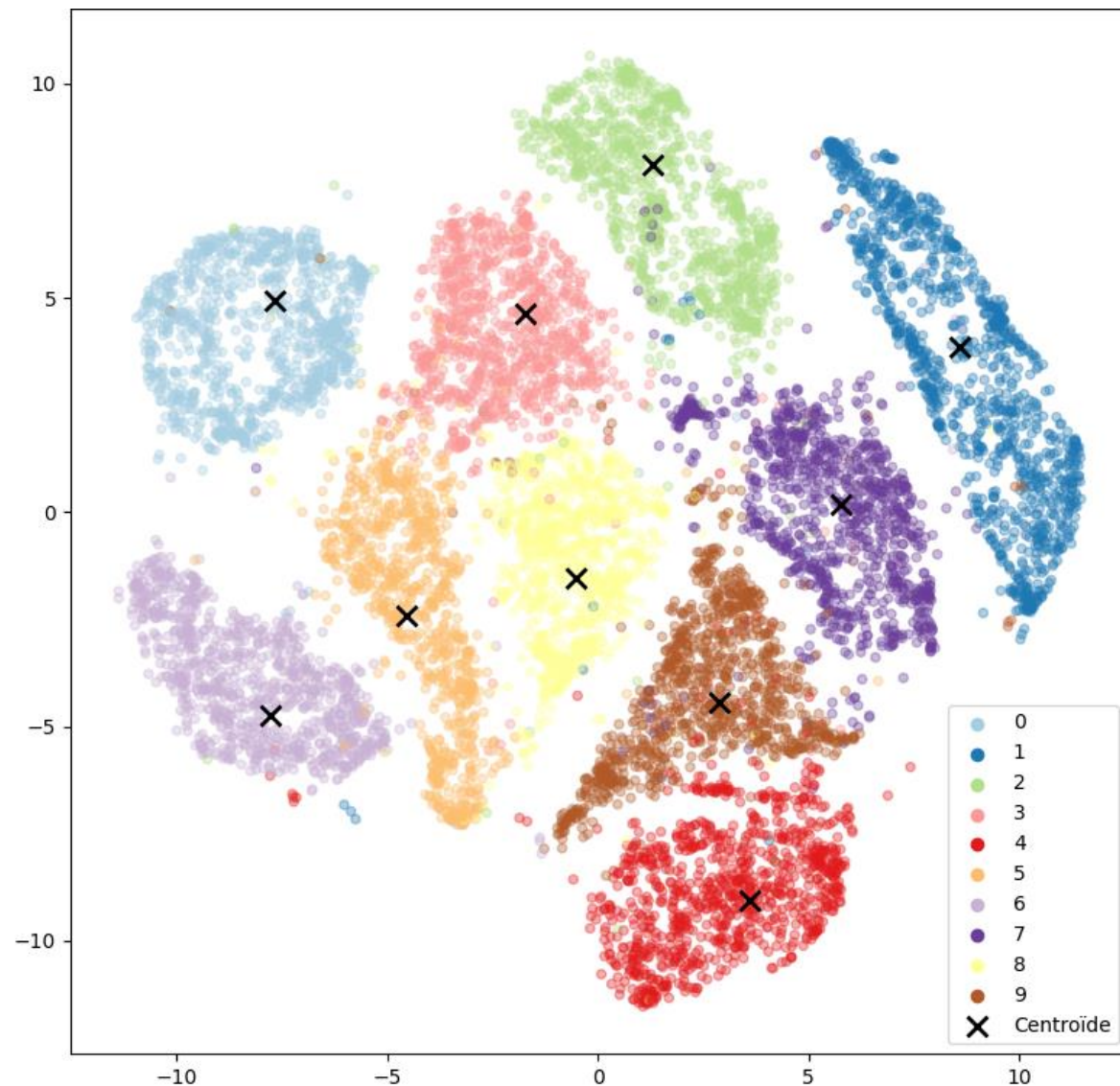
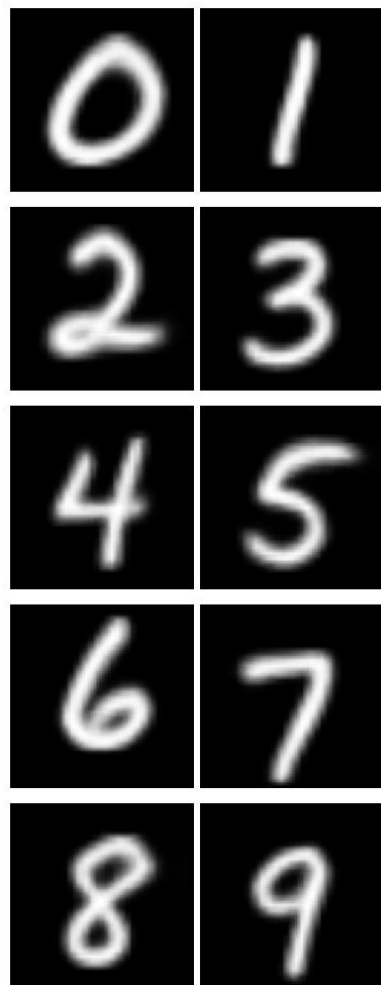
Classifieur

99,29% de précision sur la base de données MNIST

99,26% de certitude sur MNIST

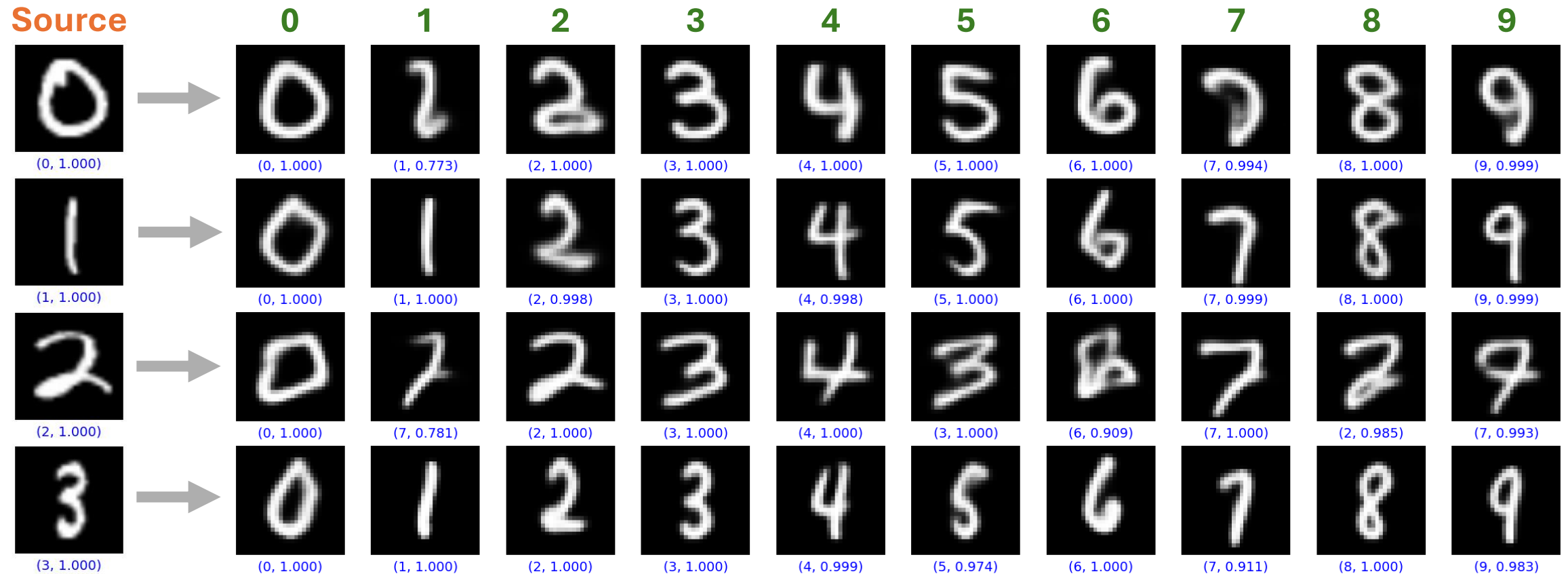
Résultats expérimentaux

Prototypes



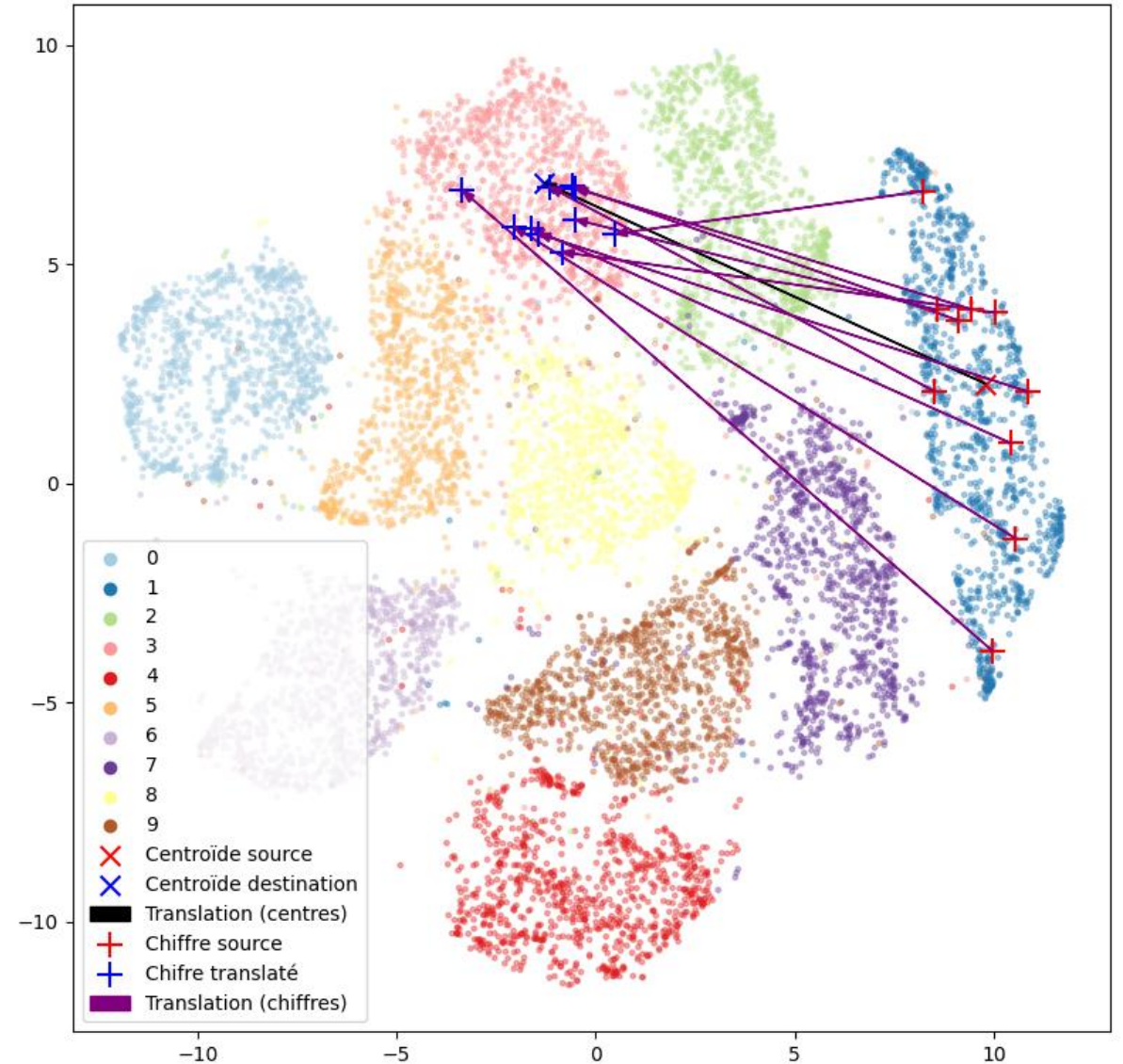
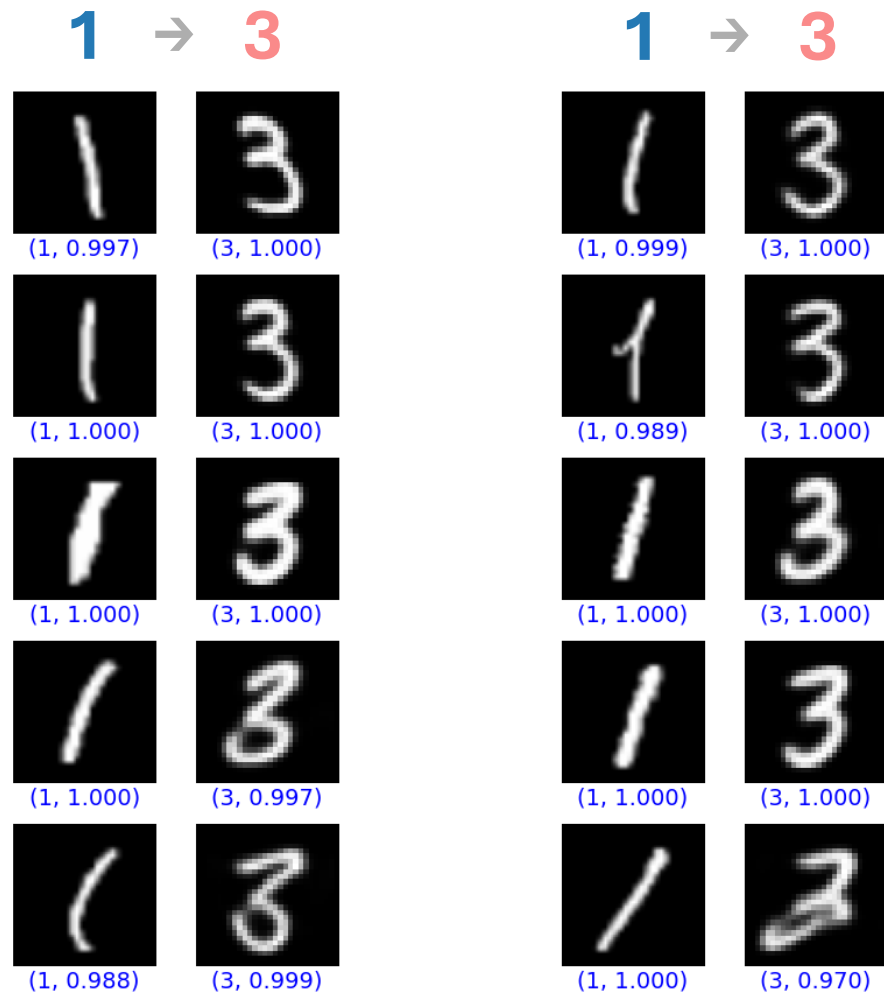
Résultats expérimentaux

Obscuration



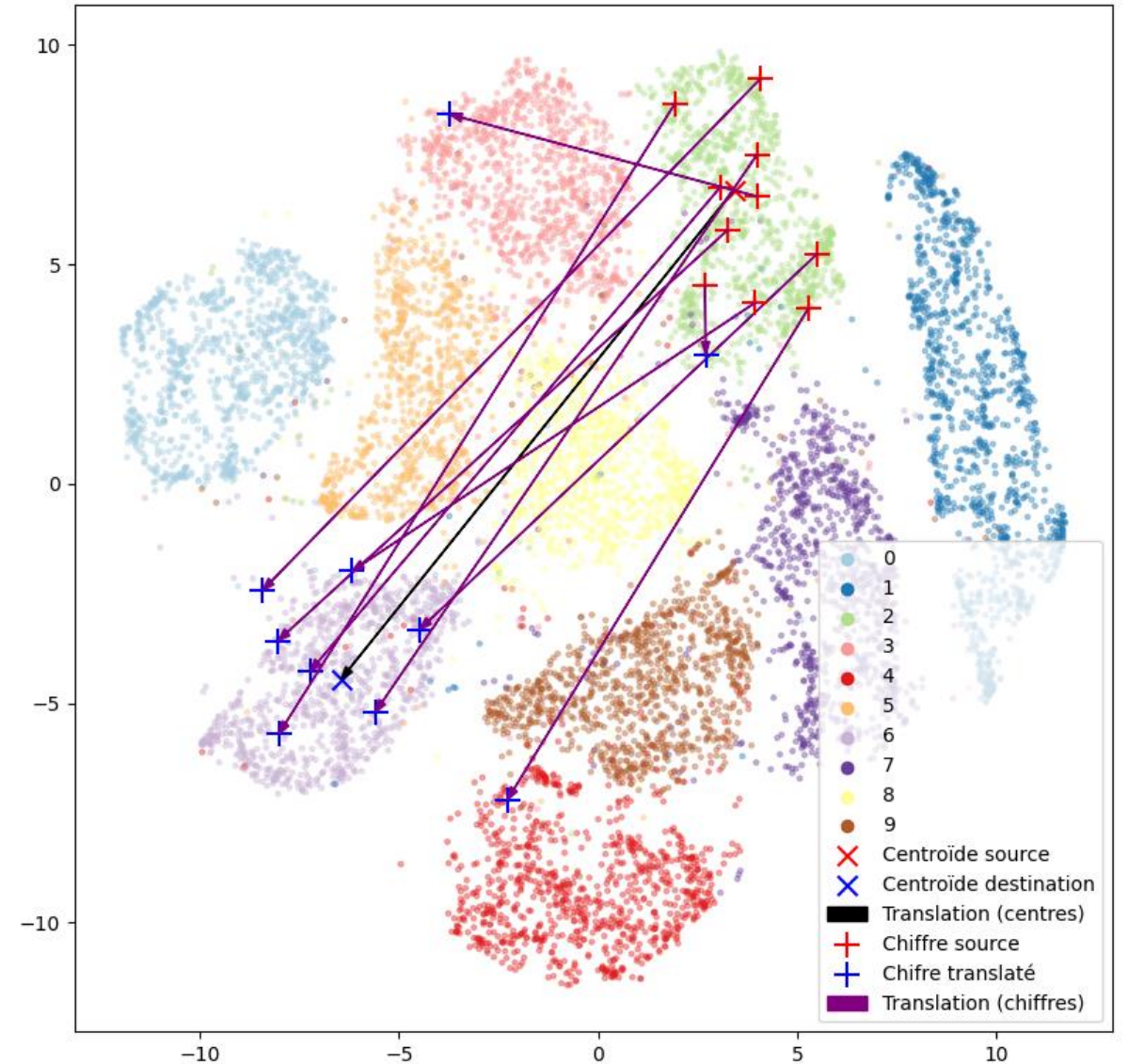
Résultats expérimentaux

Translation des chiffres



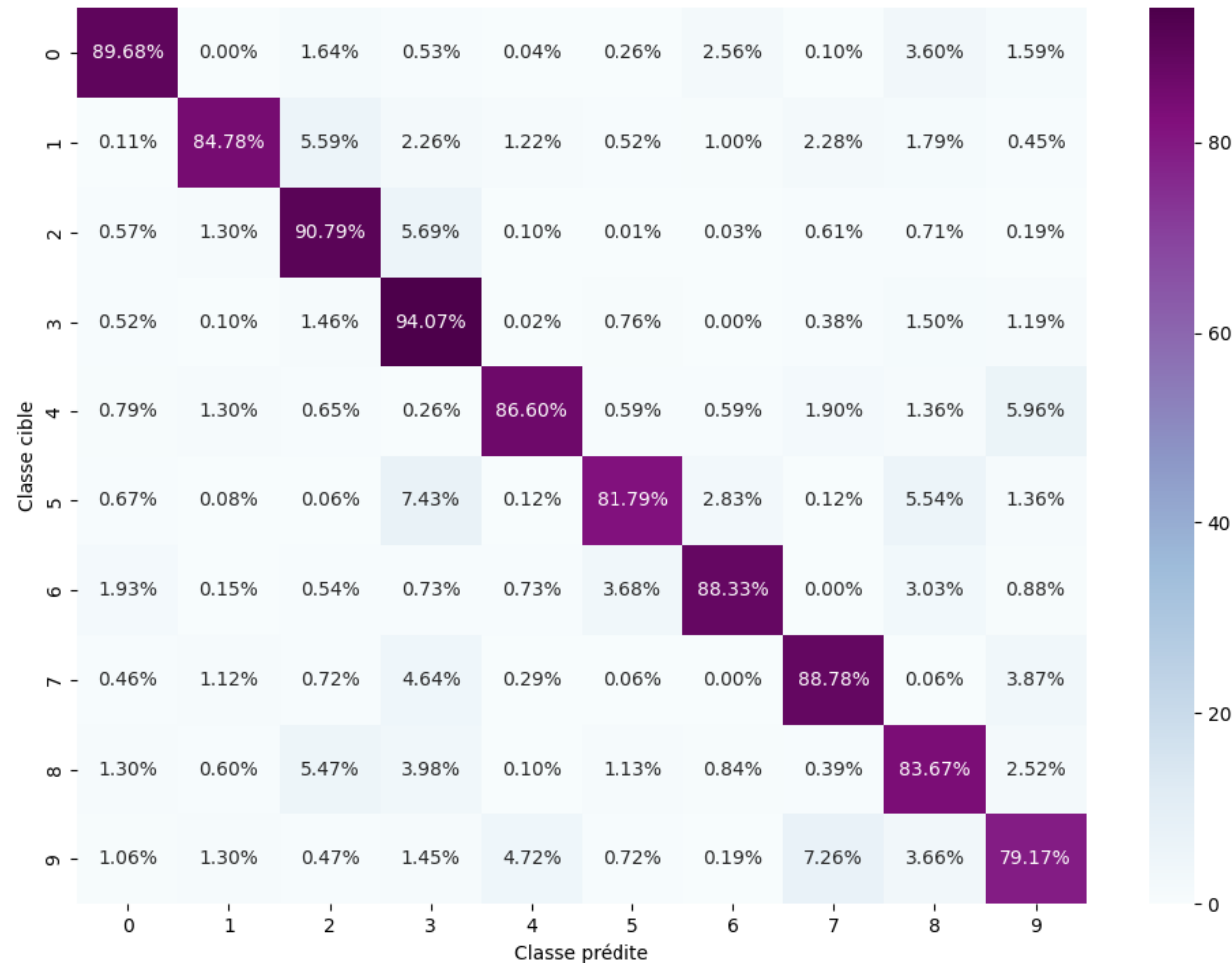
Résultats expérimentaux

Translation des chiffres



Résultats expérimentaux

Classification de l'obscuration

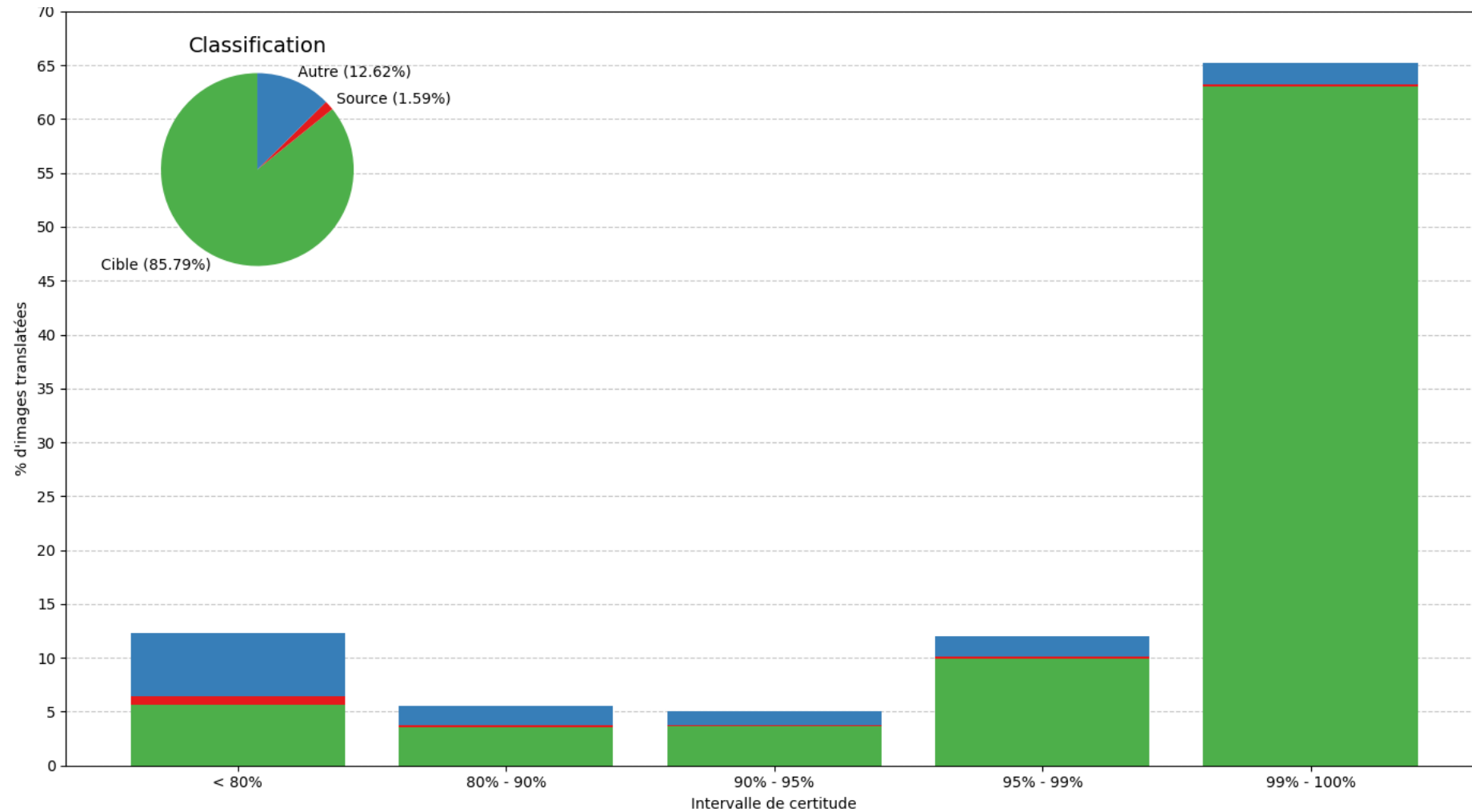


Précision : **86,77%**

Certitude : **94,17%**

Résultats expérimentaux

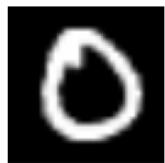
Distribution de la certitude des prédictions



Résultats expérimentaux

Désobscuration

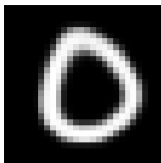
Source



(0, 1.000)

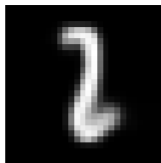


0



(0, 1.000)

1



(1, 0.773)

2



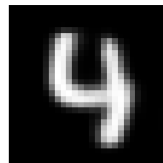
(2, 1.000)

3



(3, 1.000)

4



(4, 1.000)

5



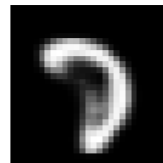
(5, 1.000)

6



(6, 1.000)

7



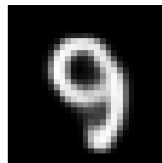
(7, 0.994)

8

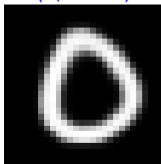


(8, 1.000)

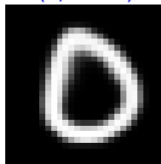
9



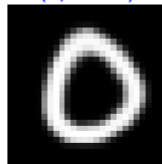
(9, 0.999)



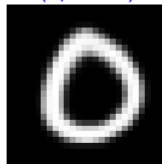
(0, 1.000)



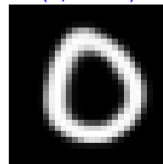
(0, 1.000)



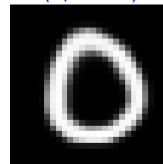
(0, 1.000)



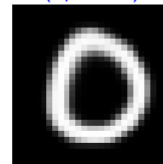
(0, 1.000)



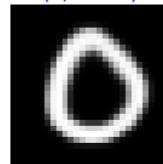
(0, 1.000)



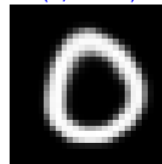
(0, 1.000)



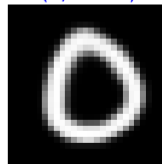
(0, 1.000)



(0, 1.000)



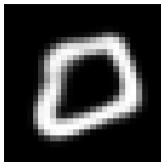
(0, 1.000)



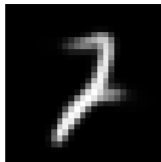
(0, 1.000)



(2, 1.000)



(0, 1.000)



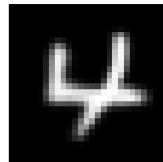
(7, 0.781)



(2, 1.000)



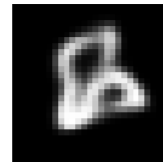
(3, 1.000)



(4, 1.000)



(3, 1.000)



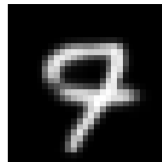
(6, 0.909)



(7, 1.000)



(2, 0.985)



(7, 0.993)



(2, 1.000)



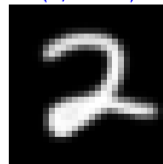
(2, 1.000)



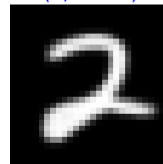
(2, 1.000)



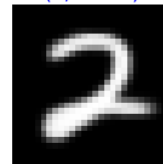
(2, 1.000)



(2, 1.000)



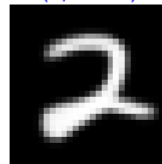
(2, 1.000)



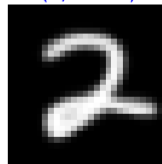
(2, 1.000)



(2, 1.000)



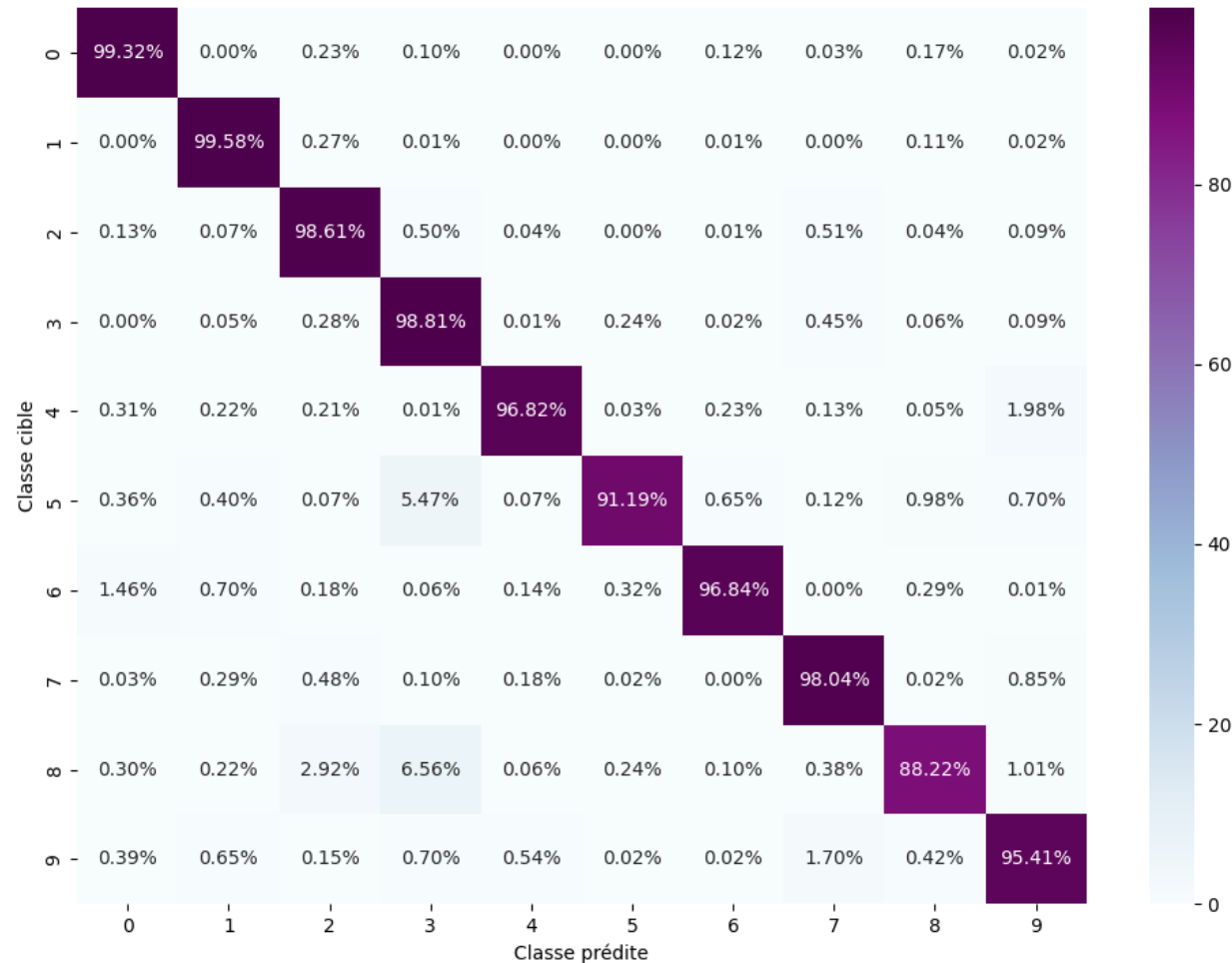
(2, 1.000)



(2, 1.000)

Résultats expérimentaux

Classification de la désobscuration



Précision : **96,41%**

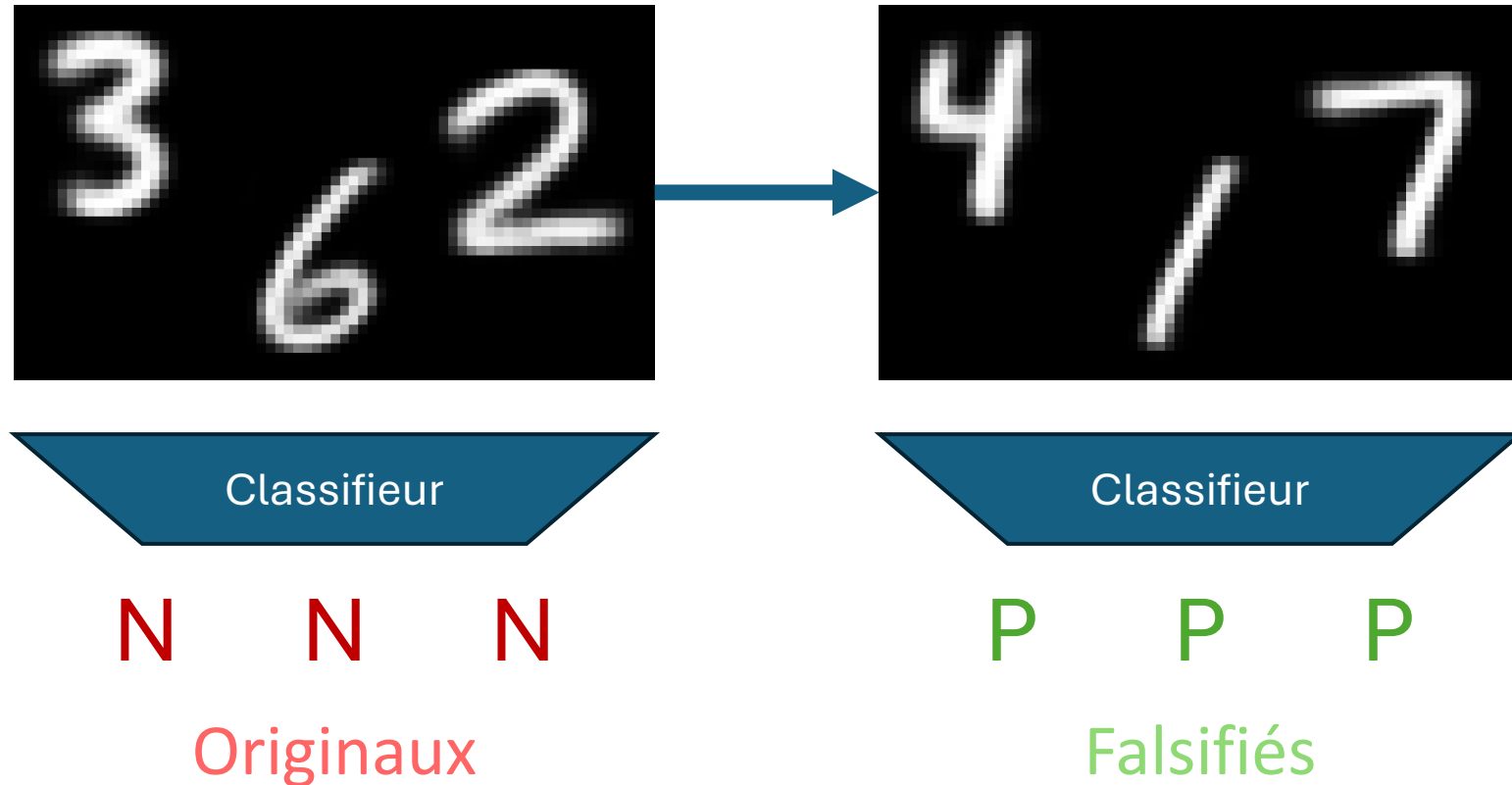
Certitude : **97,54%**

Attaque (et défense)

Détection de l'obscurisation

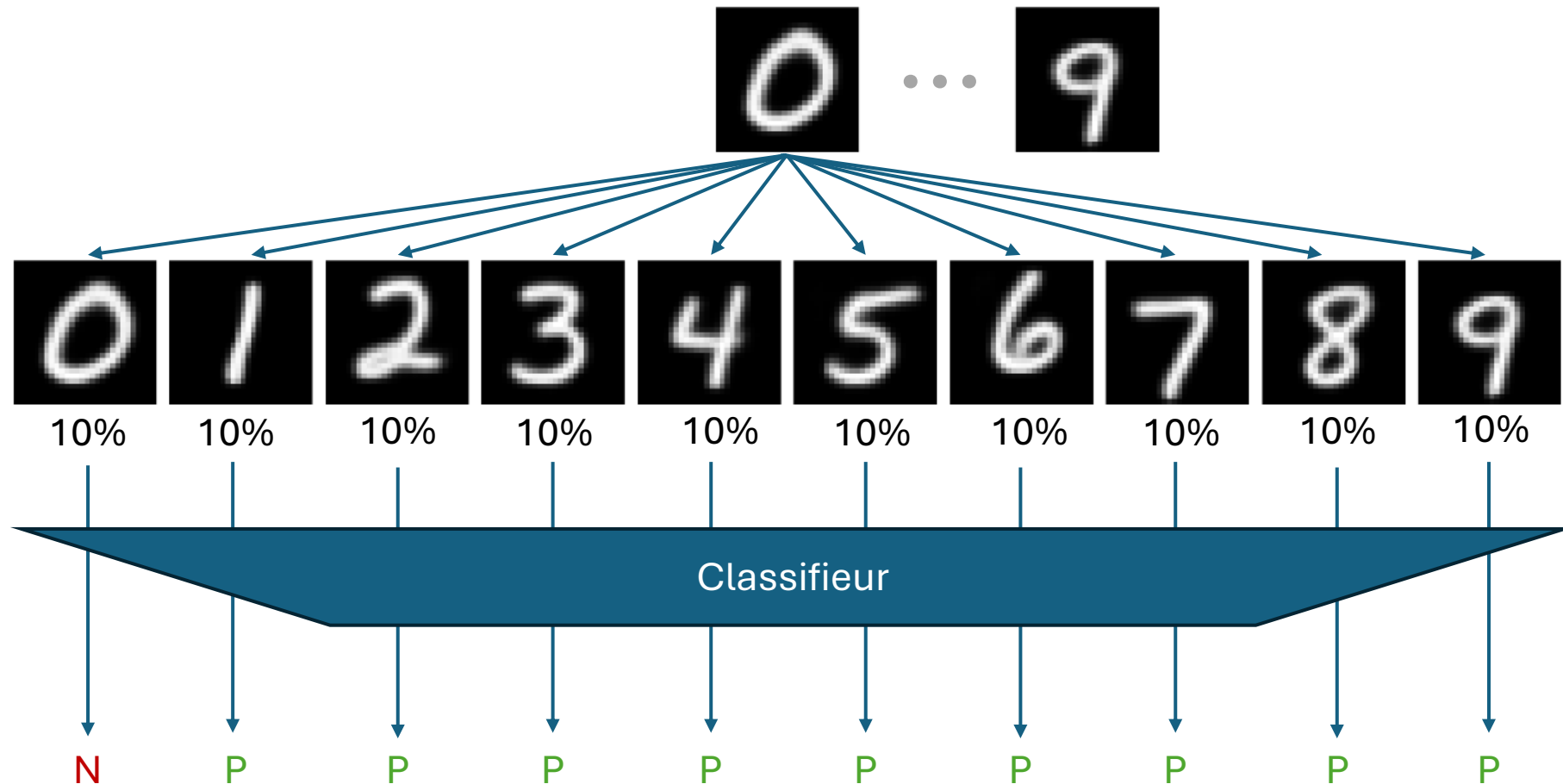
Fonctionnement

Obscuration



Détection de l'obscurité

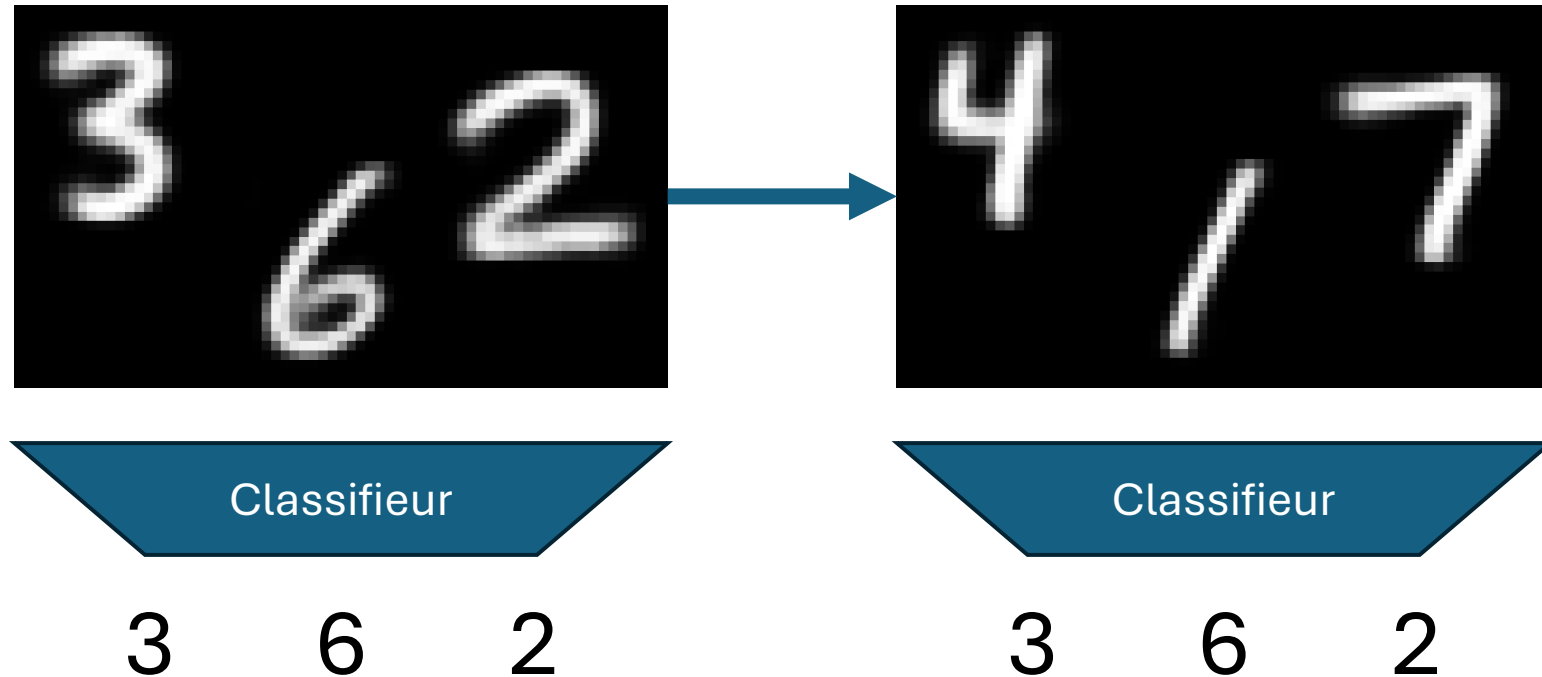
Entraînement du classifieur



Reconnaissance de la classe source (traces)

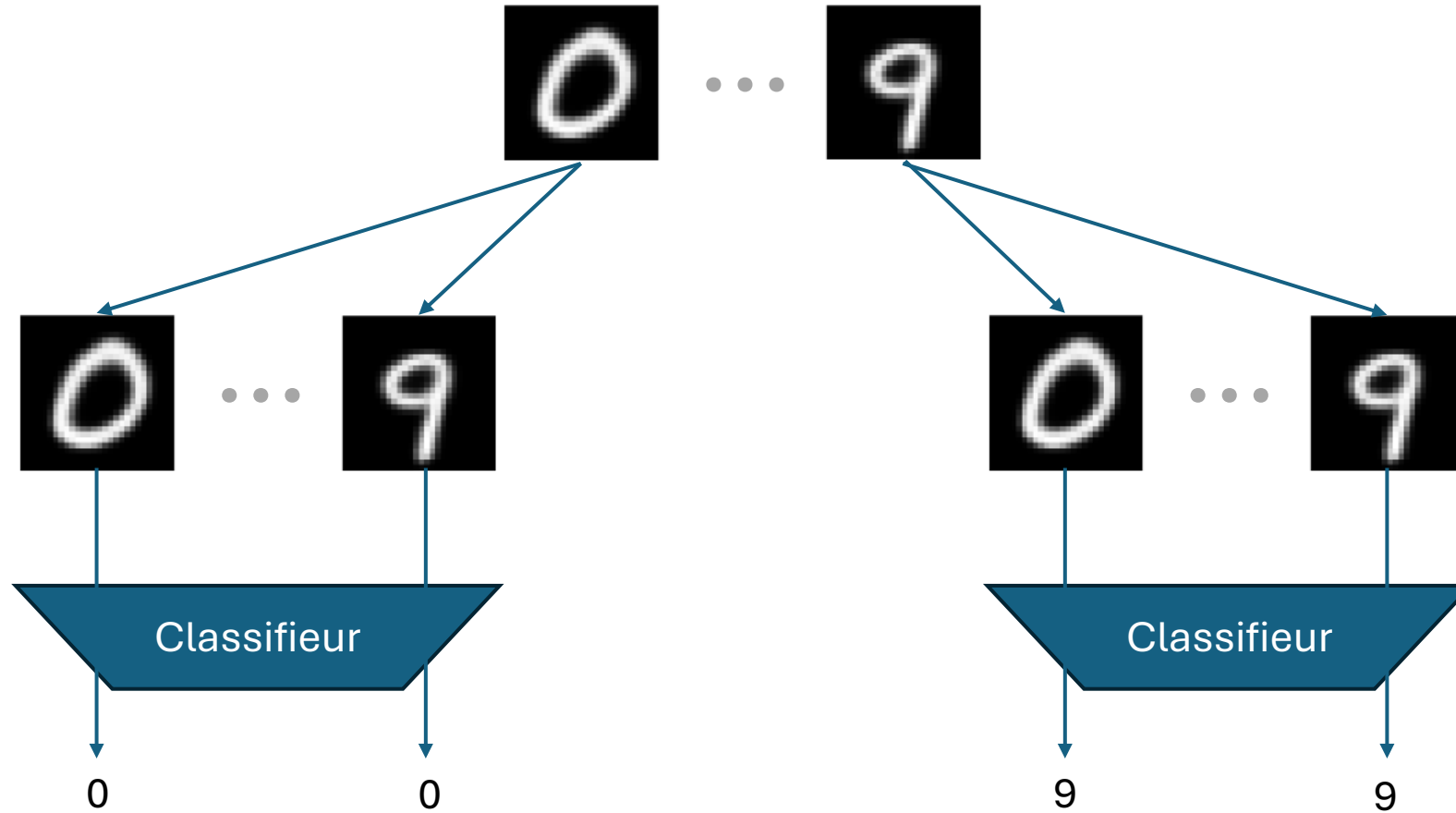
Fonctionnement

Obscuration



Reconnaissance de la classe source (traces)

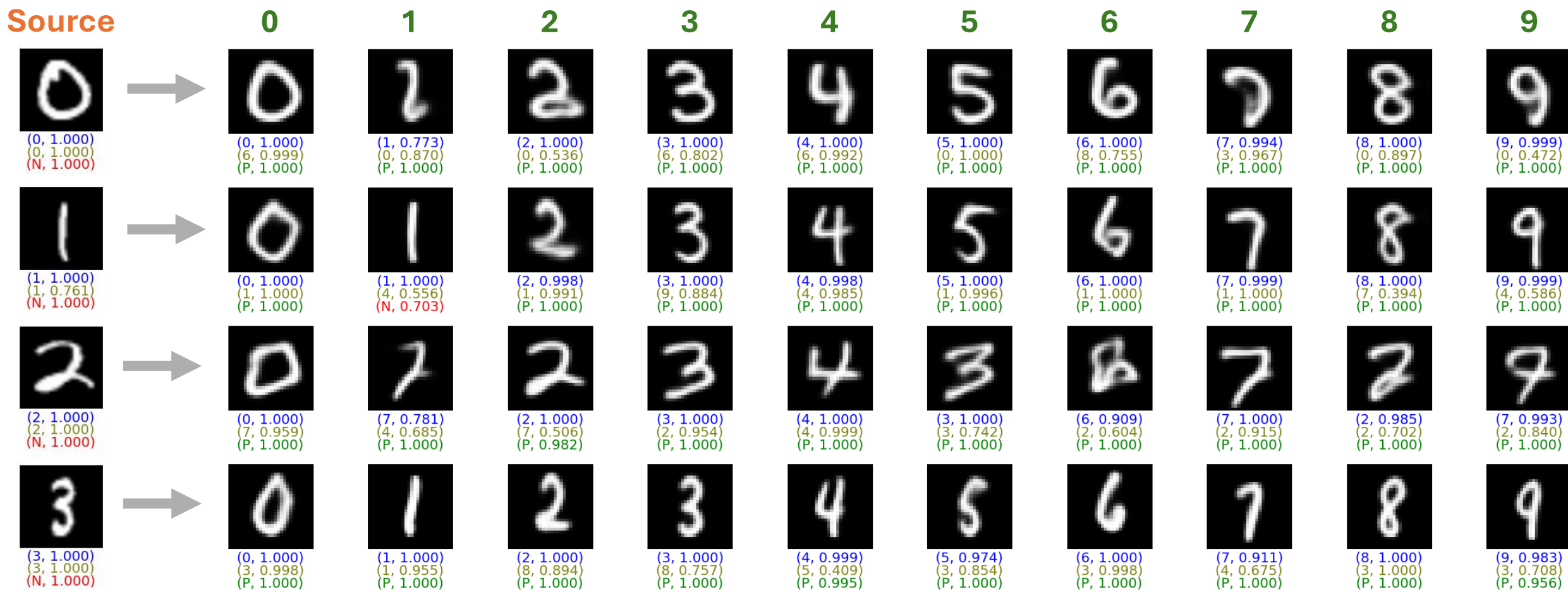
Entraînement du classifieur



Résultats expérimentaux

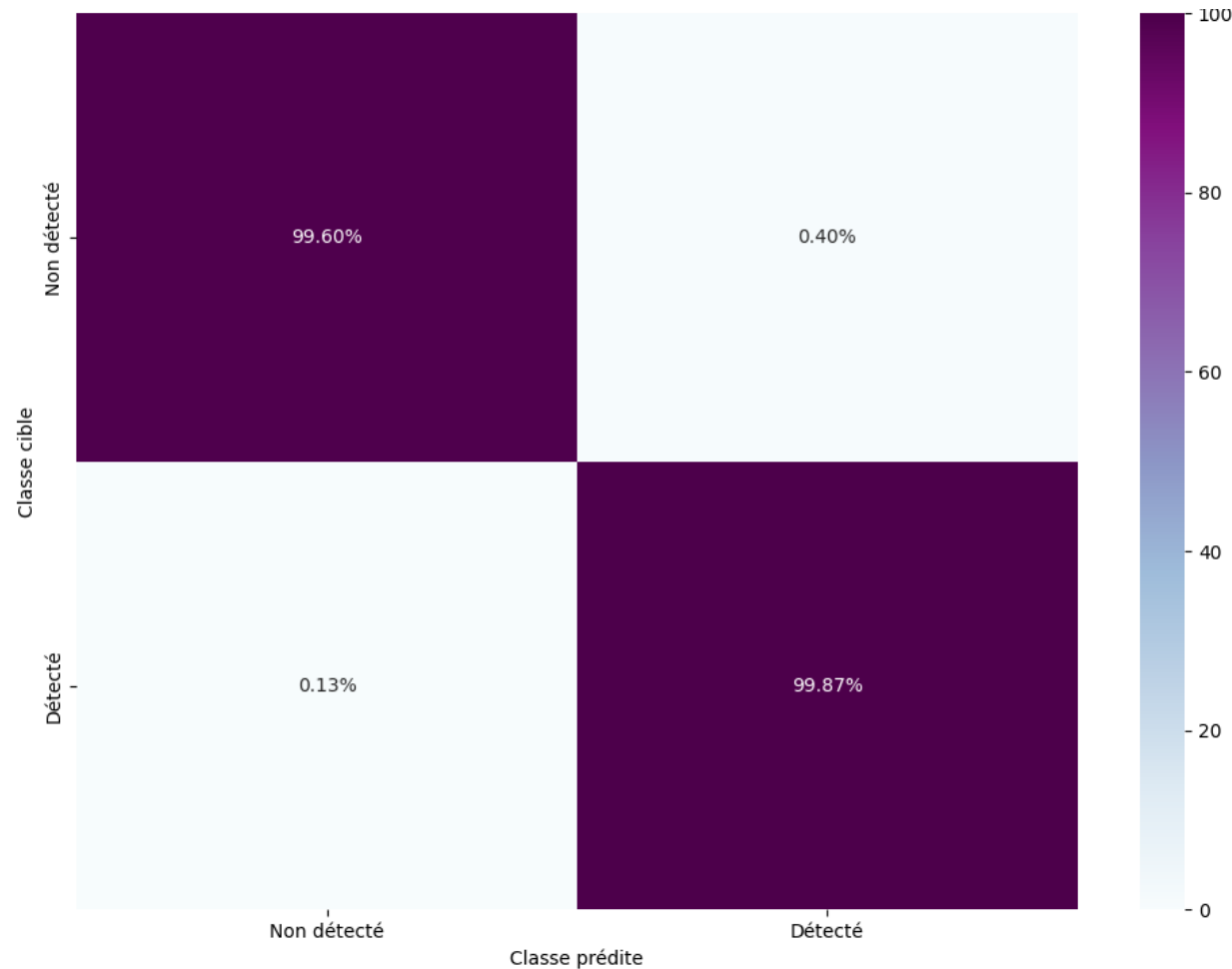
Résultats expérimentaux

Détection de l'obscurité et de la classe source



Résultats expérimentaux

Détection de l'obscurité

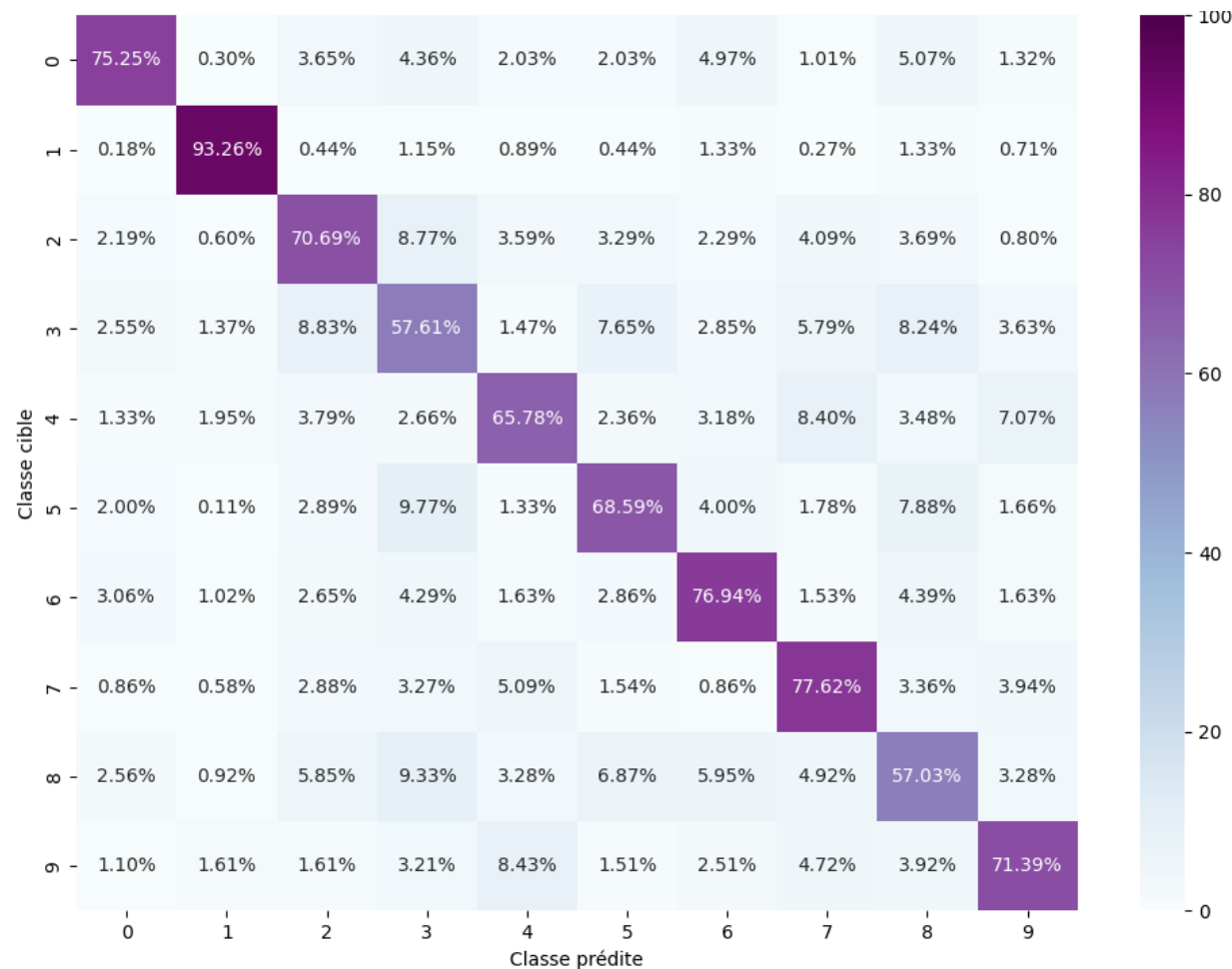


Précision : **99,84%**

Certitude : **99,95%**

Résultats expérimentaux

Reconnaissance de la classe source dans la translation (traces)



Précision : **71,75%**

Certitude : **87,24%**

Conclusion

Efficace :

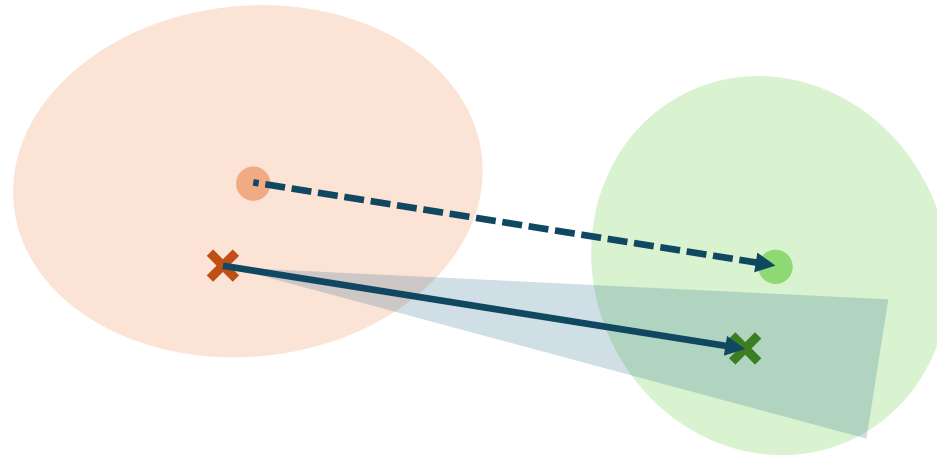
- **Obscuration** (~86% de classif. classe cible et ~12% classe source)
- **Invisible**
- **Réversible** (~94% de classif.)

Attaquable (analyse des traces) :

- **Forte reconnaissance** des traces de la classe source (~72% de classif.)
- **DéTECTABLE** (~100% de classif.)

Perspectives

- **Méthode de chiffrement** : intégration d'une clé, choix du vecteur de translation, d'un angle et d'un ratio



- **Défense** : suppression des traces dans la translation

Perspectives

- **Empoisonnement** : tromper le classifieur

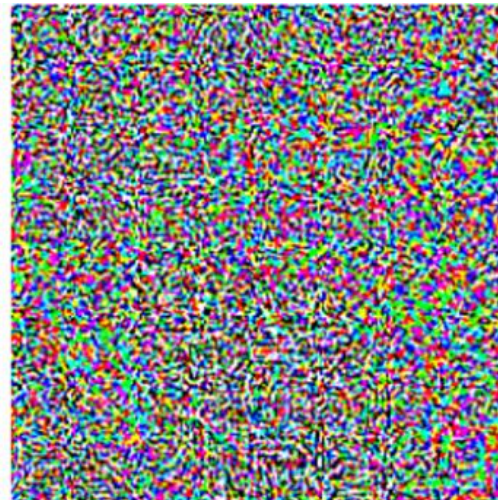


x

“panda”

57.7% confidence

$+ .007 \times$



$\text{sign}(\nabla_x \mathcal{L}_{CE}(F_\theta(x), y_{tgt}))$

“nematode”

8.2% confidence

$=$



$x +$

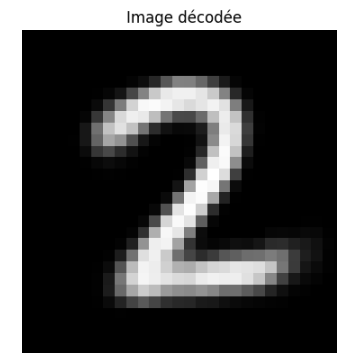
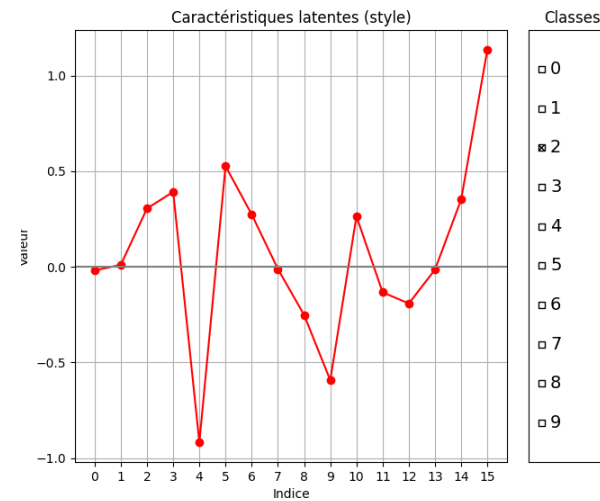
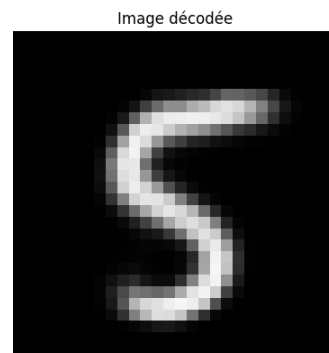
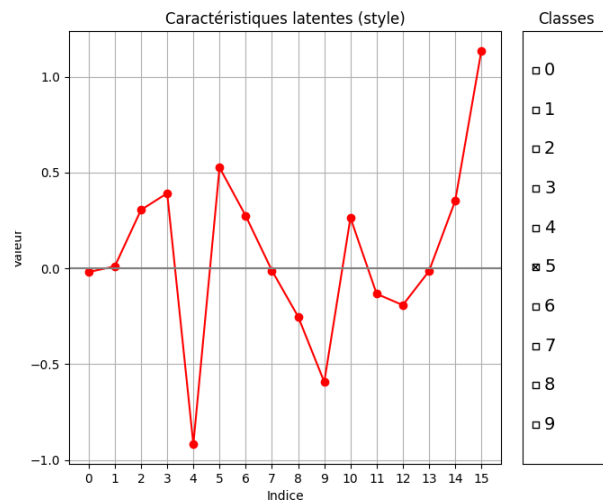
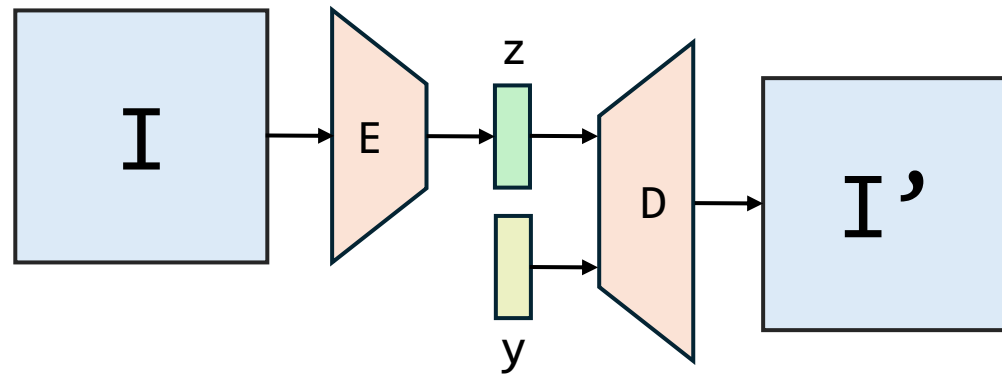
$\epsilon \text{sign}(\nabla_x \mathcal{L}_{CE}(F_\theta(x), y_{tgt}))$

“gibbon”

99.3 % confidence

Perspectives

- **CVAE** (VAE conditionnel) : découplage du style (trace?) et de la classe



Références

→ *β -VAE: Learning Basic Visual Concepts with a Constrained Variational Framework*

Irina Higgins, Loic Matthey, Arka Pal, Christopher Burgess, Xavier Glorot, Matthew Botvinick, Shakir Mohamed, Alexander Lerchner

International Conference on Learning Representations (ICLR), 2017

→ *Transferable Adversarial Patches : A Potential Threat for Real-World Computer Vision Algorithms*

Pol Labarbarie

PhD Thesis, Milad Informatique Université Paris-Saclay, 2024

→ *Learning Disentangled Representations With Semi-Supervised Deep Generative Models*

N. Siddharth, Brooks Paige, Jan-Willem van de Meent, Alban Desmaison, Noah D. Goodman, Pushmeet Kohli, Frank Wood, Philip H. S. Torr. L

2017